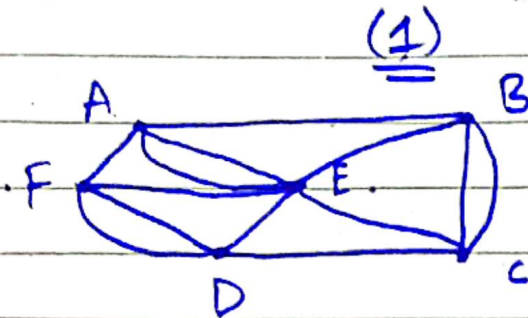


Discrete Structures

Quiz 6 (Section H, F)

Question:

Find Euler Circuit.



Yes, It is Euler Circuit because:

(1) Graph is connected.

(2) Degree of every vertex is even.

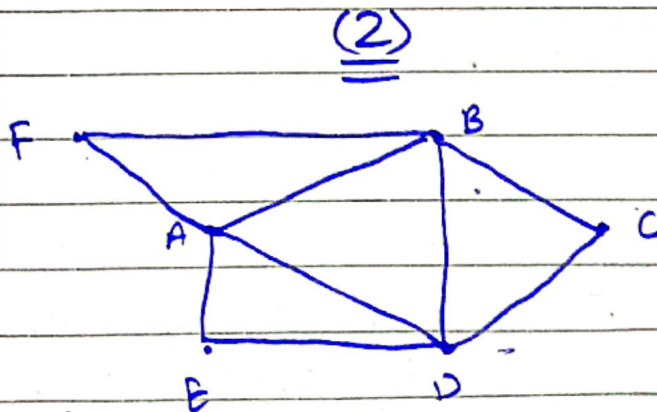
$$\deg(A)=4, \deg(B)=4, \deg(C)=4$$

$$\deg(D)=4, \deg(E)=6, \deg(F)=2$$

⇒ One valid Euler Circuit is:

A → F → D → C → B → A → E → D → ~~F~~ → ~~B~~ E

B → C → ~~A~~ → E → A



Yes, It is Euler circuit because:

(1) Graph is connected

(2) Degree of every vertex is even.

$$\deg(A)=4, \deg(B)=4, \deg(C)=2$$

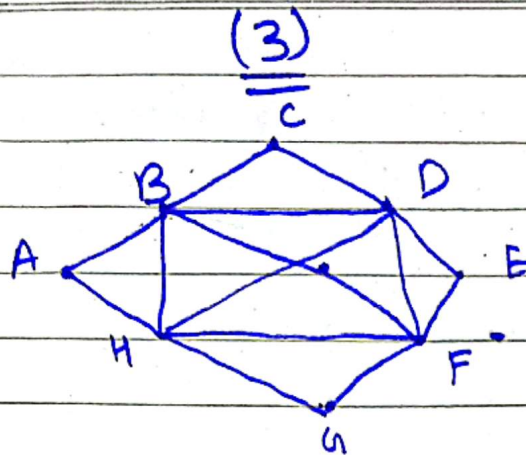
$$\deg(D)=4, \deg(E)=6, \deg(F)=2$$

⇒ One valid Euler Circuit is:

F → A → E → D → C → B → D → A → B → F

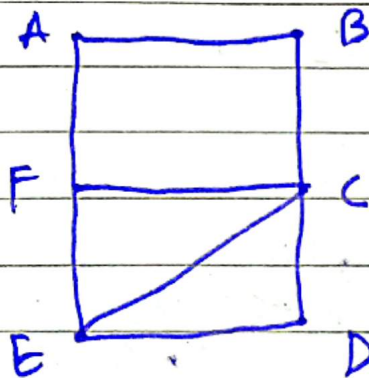
DATE: _____

DAY: _____



Not an Euler Circuit because:
 Degree of each vertex is not even.
 $\deg(B)=5$, $\deg(D)=5$, $\deg(F)=5$
 $\deg(H)=5$.

(4)



Not an Euler circuit because:
 Degree of each vertex is not even
 $\deg(F)=3$, $\deg(E)=3$